

NeuralArc Global PVT. LTD.

Custom Electronics R&D Services | IoT Product & Data analytics | Low volume manufacturing

About NeuralArc

With trust, quality, integrity and innovation as our core values, we aim to integrate harmonious partnerships with our clients. We truly believe in the transformative power of our Solution and Services to drive the right business outcome for our customers.

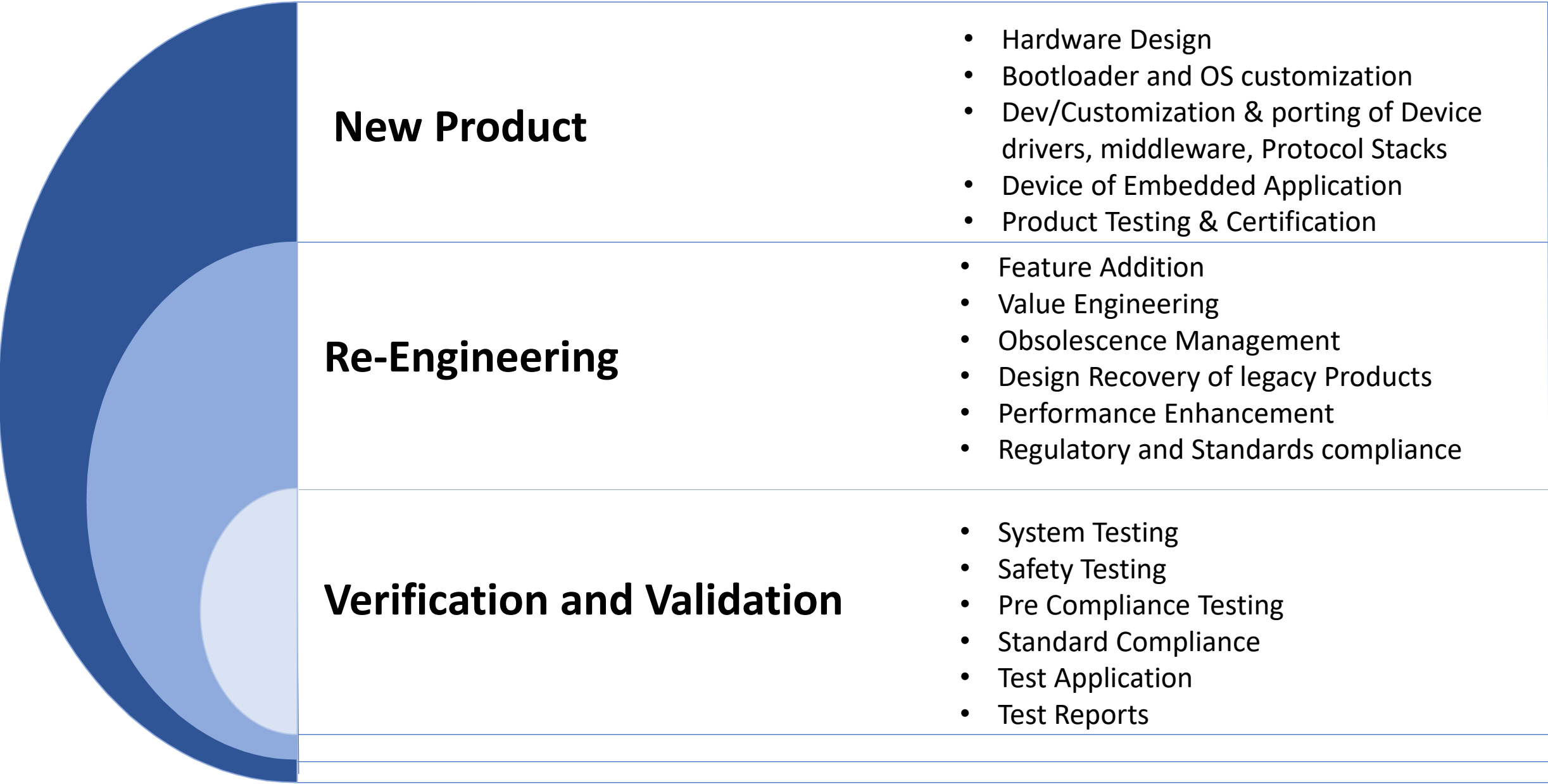
NeuralArc is a R&D firm from Coimbatore, India. A highly talented workforce of electrical and electronics engineers & data analyst provide quality service.

We are specialists in design and development of products from requirements analysis to mechanical, electrical, and software design all the way through implementation, testing, and final integration.

We offer Embedded system (including IoT) product design and development , Data Analytics, Prototyping & small volume Manufacturing Services.

Designing of systems with high quality components and optimum features to meet the expectation is our specialty





Real Time Embedded Design Capabilities



Embedded Application

- Circuit Design/PCB Layout
- Component Selection
- Digital Design & Verification
- Design Implementation and Prototyping
- BOM cost reduction
- Obsolescence removal
- Design Recovery of Legacy Products
- Performance Enhancement

Embedded Application

- Linux/Android/Microcontrollers
- MQTT/HTML/Python

BSP and Integration

- Firmware, boot loaders, Device Drivers
- Protocols
- Middleware, Protocol stacks, Drivers Customization and integration
- RTOS

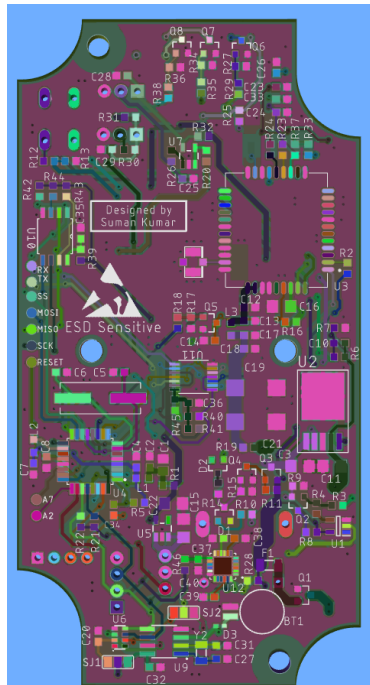
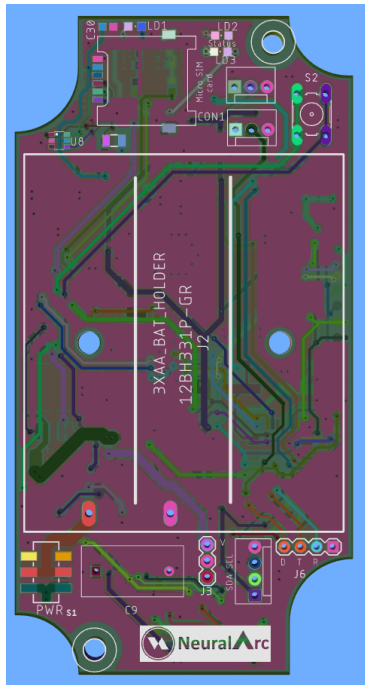
Embedded Application

- Hardware/Software Testing
- Test Automation
- Compliance Testing

High-Speed Board Design (Schematic & Layout Design)

Expertise:

- Microprocessor based design
- High-Speed PCB design involving Delay & Impedance Matching
- Multi Layer PCB Design



Service Areas :

- Schematic preparation, Component selection, BOM optimization
- PCB Design, Gerber and BOM release for fabrication and assembly
- Board Validation
 - Test Code development
 - Power-up and sanity testing of boards
- Build Jigs and Fixtures for custom board manufacturing

Case Study –Data Aggregation Device(DAD)

Single Board Computing System

A Successful commercial Deployment in USA , Spain and Vietnam

- i. Linux Computing Engine
- ii. Multi Communication interface Support
 - Ethernet WIFI, Blue tooth, LoRa and 4G
- iii. USB Host
- iv. Power Over Ethernet
- v. Local Data Storage
- vi. 8 Hours Battery Backup
- vii. Relay Control
- viii. Provision to interface more Analog/Digital Sensors



Case Study – Environmental Monitoring System

Multi Sensor Solution



**PM2.5 &
PM10 Particle
Sensor**

Pollutant Particle Sensing
Outside Air Quality Monitoring
Industry/Office/Home air quality
control

**CO, CO2,
SO2,NO, NO2**

Air Quality
Industry/Office/Home
Smart Controls

**Ethernet,
WIFI, GSM,
GPS**

Easy communication support
Multi band Network

**Smart Control
-Linux
Computing**

SSL, VPN Secured Network

**Temperature
& Humidity**

Outdoor Ambient monitoring
Industry/Office/Home

Case Study – Ventilator

Smart Ambu Bag Ventilator

- i. Integrated with multiple sensors:
 - Blood oxygen level Sensor.
 - Pressure Sensor
 - Air Temperature & Humidity Sensor.
 - Air Flow Rate Sensor
- ii. Sharp & clear display with 3 Buttons user interface
- iii. Manual and Patient Trigger support
- iv. Timer based operation
- v. Programmable operation for adult and infant.
- vi. 24 Hours Battery Backup



Product Portfolio



- Smart Agriculture



- Industrial Automation



- Medical

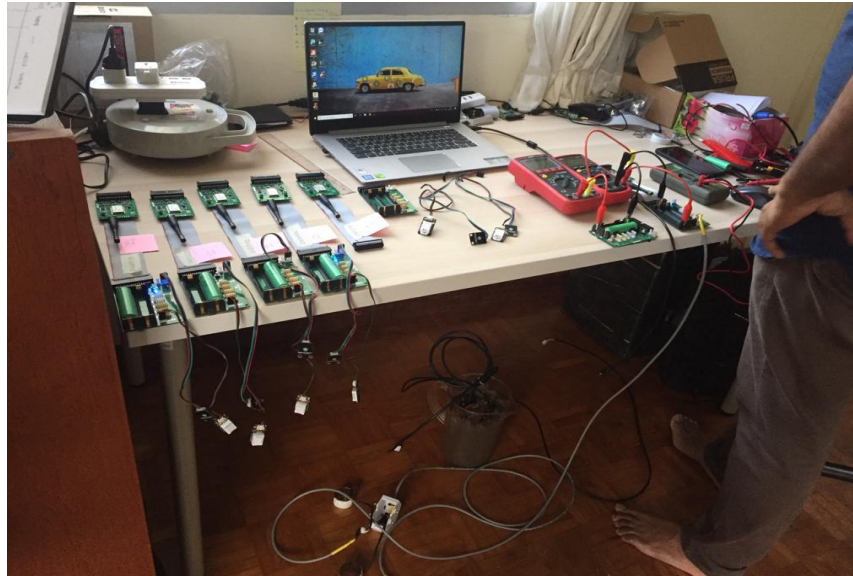


- Commercial

Manufacturing Support System

Tie-Up With Multiple EMS Companies

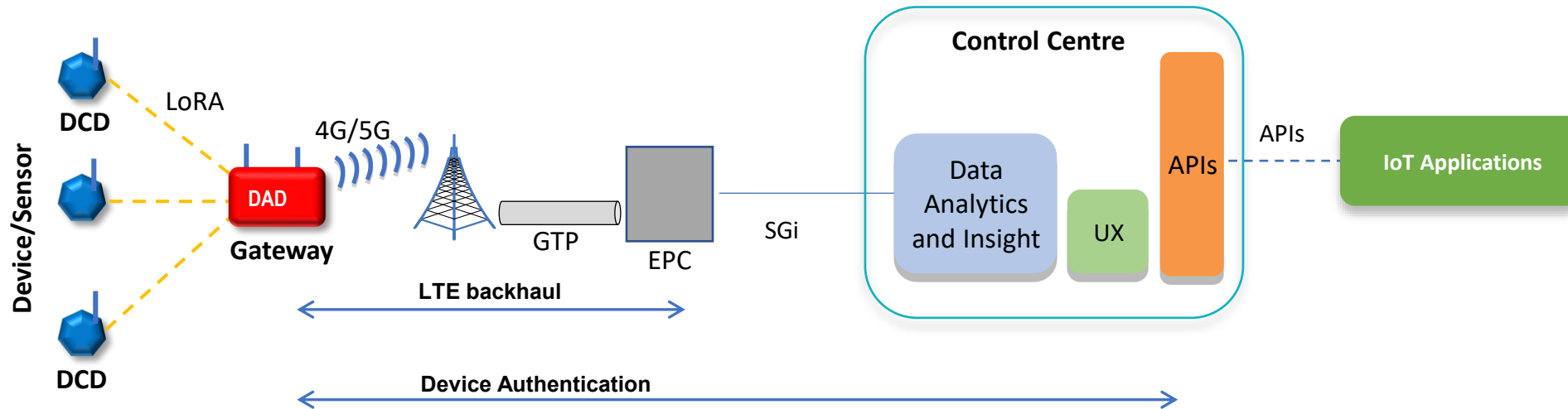
- MightyNet – **Taiwan**
- 3D –Technologies PVT. LTD
- **Bangalore**
- Vasantha Advanced Systems
- **Coimbatore**



Smart Agriculture Solution

NeuralArc

Solution Architecture



We supply two device in the proposed architecture

1. Gateway (Data Aggregation Device (DAD))

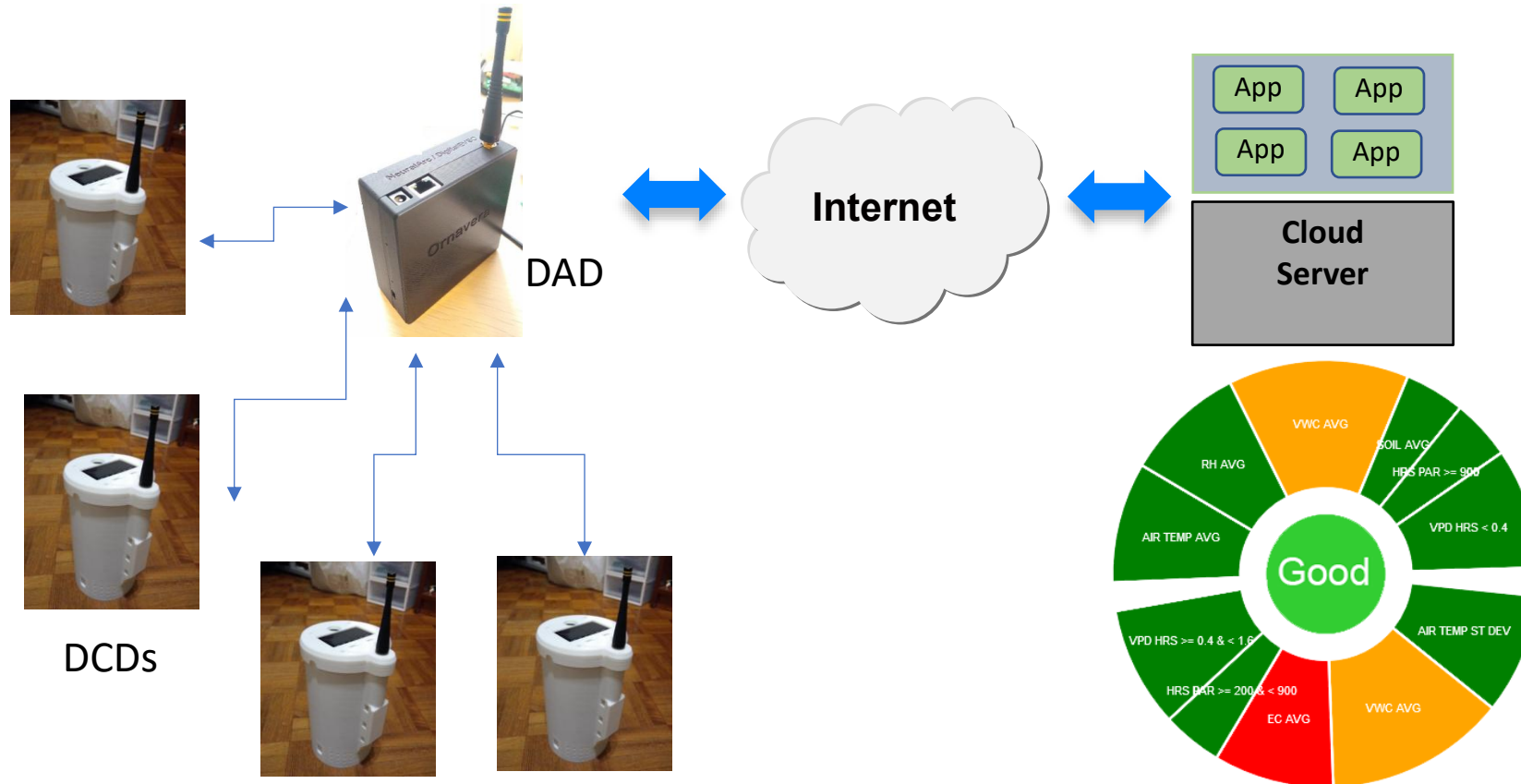
- Efficient and fast compute engine
- Integrated secured multi communication network support like 4G/5G Cellular and WIFI.
- Long Range local communication network LoRA with multiple sensor nodes
- Variety of Interface buses
- Interactive display for multi use
- Wide Operating Electrical Condition

2. Sensor node (Data Collection Device (DCD))

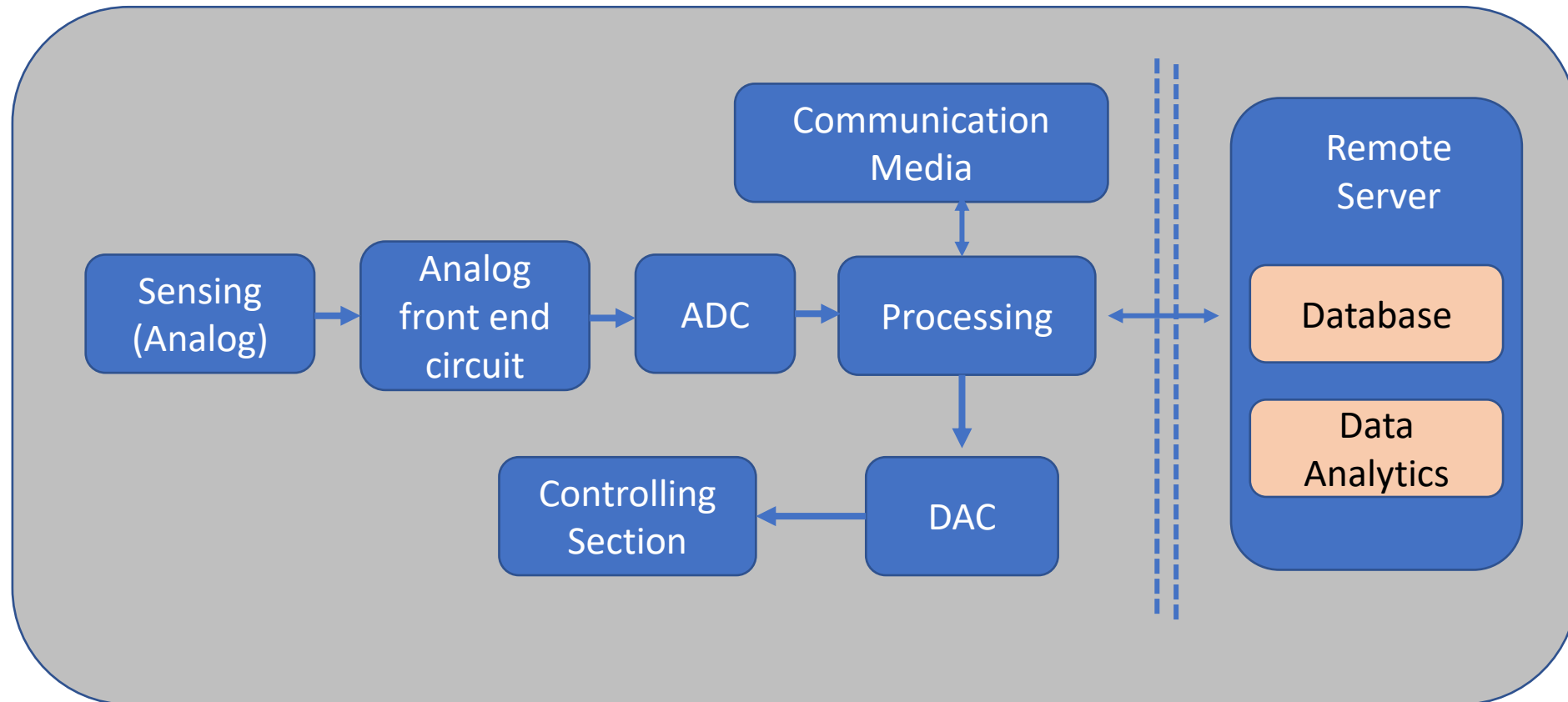
- Micro controller smart engine
- Multiple Sensor Interfaces (Digital/Analog)
- Long Range local communication network LoRA
- Interactive display for multi use

Smart Agriculture Network

Smart Agri Solution is designed to help growers improve crop health by providing in-depth information in a simple platform. This solution eliminates the hassle of manually calculating data and provides growers with detailed crop information and the opportunity to automate the whole operation and generate useful insights.



IoT based typical Sensor Node



Features

- Efficacy of Ornavera sensors for all measurements: Climate, Soil and Plant level.
 - Temp and Humidity +/- 2%
 - Lux/PPFD: +/- 10%
 - Soil:
 - Capture readings from up to 2 soil sensors per DCD
 - Temp to +/- 2 degrees
 - Permittivity/VWC: +/- 5% (with soil calibration)
 - EC: +/- 5% (with soil calibration)
 - Plant stem diameter:
 - Capture readings from up to 2 dendrometers per DCD
 - Accuracy to +/- 0.05mm
- Data capture and upload:
 - Ease of installation
 - Minimal DCD maintenance
 - Battery life
 - Stability of sensors on mounting media (wooden/metal poles)
 - Broad LoRa coverage footprint: 15km radius
 - Low error rate of message send/receive (< 1%)
 - DAD reliability
 - Low maintenance (no need to reboot manually)
 - Battery backup: > 24 hours
 - Secure data transmission & storage

Application SOFTWARE

- Effective data capture/process/storage
 - Capture/store raw measurements
 - Organise by time/device/location
 - Accurate derivations
 - VWC from permittivity
 - Pore water EC from bulk soil EC
 - Stem diameter from mV
 - PPFD from Lux
 - VPD (saturation)
 - PAR hours
 - DLI
 - Other mathematical calculations (High/Low/Average/Std Deviation)
- Login & credentials
 - Secure
 - Different levels of access (admin & standard user)
 - Configurable user preferences
 - Ability to arrange view of sensors as they are laid out spatially on the farm
- Simple, accurate visual insights
 - Live data Dashboards
 - 14 day summary graphical displays
 - Ornavera Eye
 - Crop health status
 - Hazard (disease) status – specifically Phytophthora
 - Analytics
 - Graphing of live & summary data
 - Download of live & summary data
- Easy to setup
 - Facilities, users, locations, devices, soil types
 - Event definitions
 - Live data & CHAP filters

Data Collection Device(DCD)

DCD is a sensor node that hosts many sensors and perform the measurement of Luminous, Atmosphere Temperature and Humidity, Soil Temperature and Humidity and Soil Electrical conductivity. These measured data are sent to the Data Aggregator Device(DAD) periodically over LoRa RF interface. The LoRa RF range is around 15 Kms

FUNCTIONAL SPECIFICATIONS

LoRa:

- FCC compliant
- International capability

I/O:

- Modular/Plug & Play
- SPI (Dendrometer - Sensor) X 2
- RS485 (Soil sensors) X 2
- I2C (Temp/humidity, Lux sensor) X 4
- Simplified connectors to maximize durability

Power management

- Maximize battery life
- Eliminate power waste from passive components
- Shut off power when in sleep mode

Solar Charger:

- Healthy SV power from solar panels

CPU

- Reliable
- Fast
- Simple / low power



Data Aggregation Device(DAD) - Gateway

DAD collects sensor data from multiple Data Collection Devices(DCD) over LoRa(686 Mhz to 928 Mhz) RF interface and store it in the local SD card. Besides, sends the collected data periodically to the cloud server over SSL. Multiple communication interface to the cloud server - Ethernet, WiFi and LTE.

FUNCTIONAL SPECIFICATIONS

LoRa :

- Sensor Data collection interface
- FCC compliant
- International capability

Could Interface : Ethernet/WiFi/LTE

Security :

- SSL
- MQTT protocol (password)

Local Storage :

- SSD - 32 MB

Over the Air (OTA) Firmware upgrades

Power over Ethernet

UL/CE approved

Battery backup (DC Charger)

LED status info



Field Trial

Reference Field Deployments



4G backhaul



Olive Plant in Spain

Cannabis Plantation at Colorado, USA

Pepper Plantation in Vietnam



Trial Site - Vietnam

Olam Pepper Plantation - Vietnam



- The blue forty-foot container (FFC) near the blue dot is where DAD located.
 - There are two greenhouse nurseries (another one just below the one in picture – same size)
 - Plantation extends 3.2km N and 2km E from FCC.
 - We ran an external antenna on top of the FCC.
- Plots (red dots) are .7 – 1.4 Ha Currently 150+ plots planted
 - Ultimately will have 300+ DCDs in place for final deployment
 - Poles extend 3.4m from soil surface.

DCD installation

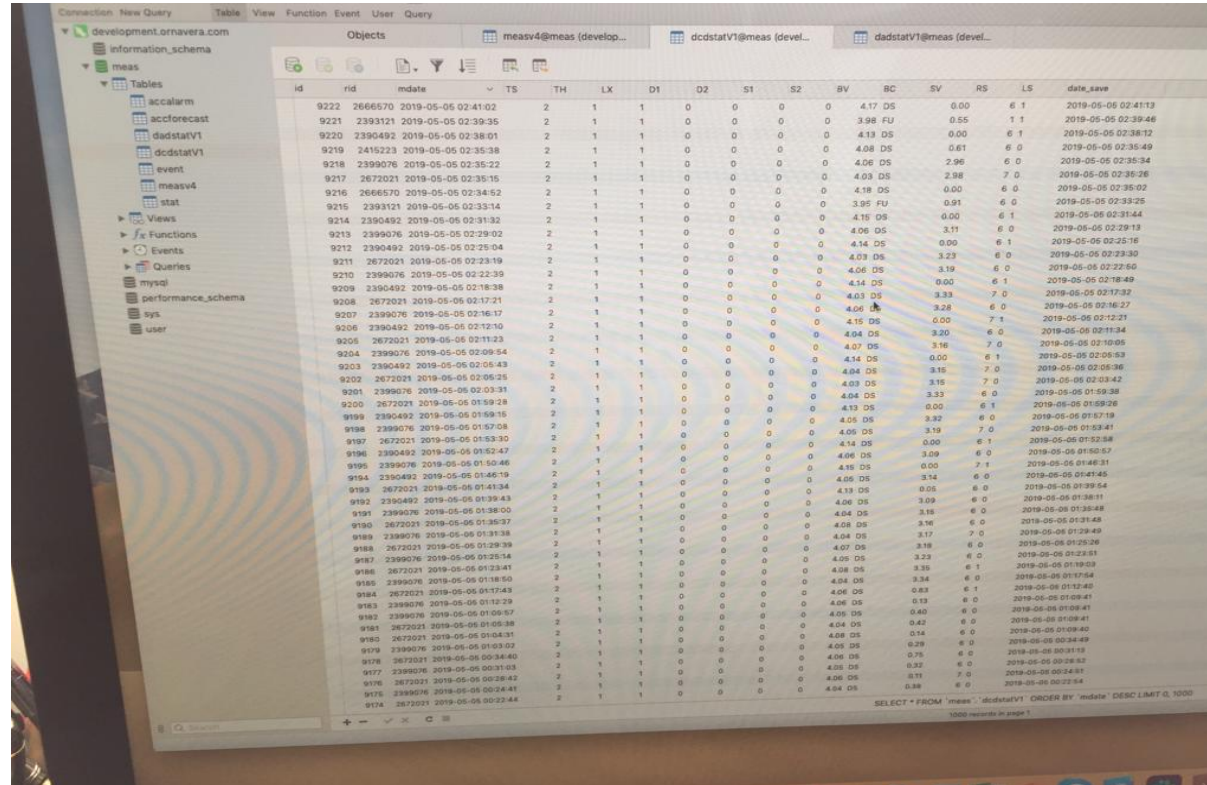


Dendrometer Installation



Dendrometer

Real-Time Data collection



The screenshot shows a MySQL database interface with a table named 'measv4' in the 'meas' database. The table contains sensor data with columns: id, rid, mdate, TS, TH, LX, D1, D2, S1, S2, BV, BC, SV, RS, LS, and date_save. The data is displayed in a grid format, showing multiple rows of sensor readings. The interface includes a sidebar with a tree view of the database structure, including tables, views, functions, queries, and schemas.

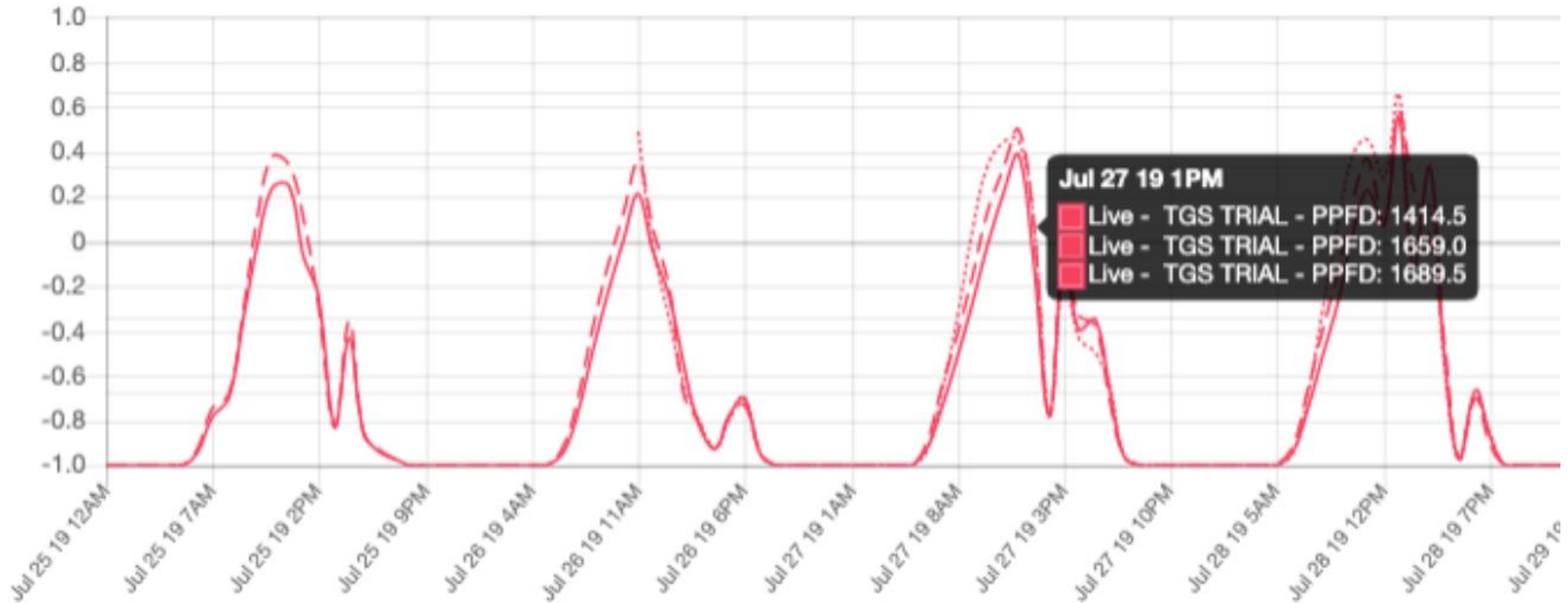
id	rid	mdate	TS	TH	LX	D1	D2	S1	S2	BV	BC	SV	RS	LS	date_save
9222	2666570	2019-05-05 02:41:02	2	1	1	0	0	0	0	4.17	DS	0.00	6	1	2019-05-05 02:41:13
9221	2393121	2019-05-05 02:39:35	2	1	1	0	0	0	0	3.98	FU	0.55	1	1	2019-05-05 02:39:46
9220	2390492	2019-05-05 02:38:01	2	1	1	0	0	0	0	4.13	DS	0.00	6	1	2019-05-05 02:38:12
9219	2415223	2019-05-05 02:35:38	2	1	1	0	0	0	0	4.08	DS	0.61	6	0	2019-05-05 02:35:49
9218	2399076	2019-05-05 02:35:22	2	1	1	0	0	0	0	4.06	DS	2.96	6	0	2019-05-05 02:35:34
9217	2672021	2019-05-05 02:35:15	2	1	1	0	0	0	0	4.18	DS	0.00	7	0	2019-05-05 02:35:26
9216	2666570	2019-05-05 02:34:52	2	1	1	0	0	0	0	4.03	DS	0.00	6	0	2019-05-05 02:35:02
9215	2393121	2019-05-05 02:33:14	2	1	1	0	0	0	0	3.95	FU	0.91	6	0	2019-05-05 02:33:25
9214	2390492	2019-05-05 02:31:32	2	1	1	0	0	0	0	4.15	DS	0.00	6	1	2019-05-05 02:31:44
9213	2399076	2019-05-05 02:29:02	2	1	1	0	0	0	0	4.06	DS	3.11	6	0	2019-05-05 02:29:13
9212	2390492	2019-05-05 02:25:04	2	1	1	0	0	0	0	4.14	DS	0.00	6	1	2019-05-05 02:25:16
9211	2672021	2019-05-05 02:23:19	2	1	1	0	0	0	0	4.06	DS	3.19	6	0	2019-05-05 02:23:30
9210	2399076	2019-05-05 02:22:39	2	1	1	0	0	0	0	4.14	DS	0.00	6	1	2019-05-05 02:22:50
9209	2390492	2019-05-05 02:18:38	2	1	1	0	0	0	0	4.03	DS	3.33	7	0	2019-05-05 02:18:49
9208	2672021	2019-05-05 02:17:21	2	1	1	0	0	0	0	4.06	DS	3.28	6	0	2019-05-05 02:17:32
9207	2399076	2019-05-05 02:16:17	2	1	1	0	0	0	0	4.15	DS	0.00	7	1	2019-05-05 02:16:27
9206	2390492	2019-05-05 02:12:10	2	1	1	0	0	0	0	4.04	DS	3.20	6	0	2019-05-05 02:12:21
9205	2672021	2019-05-05 02:11:23	2	1	1	0	0	0	0	4.07	DS	3.16	7	0	2019-05-05 02:11:34
9204	2399076	2019-05-05 02:09:54	2	1	1	0	0	0	0	4.14	DS	0.00	6	1	2019-05-05 02:10:05
9203	2390492	2019-05-05 02:05:43	2	1	1	0	0	0	0	4.04	DS	3.16	7	0	2019-05-05 02:05:53
9202	2672021	2019-05-05 02:05:25	2	1	1	0	0	0	0	4.03	DS	3.15	7	0	2019-05-05 02:05:36
9201	2399076	2019-05-05 02:03:31	2	1	1	0	0	0	0	4.04	DS	3.33	6	0	2019-05-05 02:03:42
9200	2672021	2019-05-05 01:59:28	2	1	1	0	0	0	0	4.13	DS	0.00	6	1	2019-05-05 01:59:38
9199	2390492	2019-05-05 01:59:15	2	1	1	0	0	0	0	4.05	DS	3.32	6	0	2019-05-05 01:59:26
9198	2399076	2019-05-05 01:57:08	2	1	1	0	0	0	0	4.05	DS	3.19	7	0	2019-05-05 01:57:19
9197	2672021	2019-05-05 01:53:30	2	1	1	0	0	0	0	4.14	DS	0.00	6	1	2019-05-05 01:53:41
9196	2390492	2019-05-05 01:52:47	2	1	1	0	0	0	0	4.06	DS	3.09	6	0	2019-05-05 01:52:58
9195	2399076	2019-05-05 01:50:46	2	1	1	0	0	0	0	4.15	DS	0.00	7	1	2019-05-05 01:50:57
9194	2390492	2019-05-05 01:46:19	2	1	1	0	0	0	0	4.06	DS	3.14	6	0	2019-05-05 01:46:30
9193	2672021	2019-05-05 01:41:34	2	1	1	0	0	0	0	4.13	DS	0.00	6	0	2019-05-05 01:41:45
9192	2390492	2019-05-05 01:39:43	2	1	1	0	0	0	0	4.06	DS	3.09	6	0	2019-05-05 01:39:54
9191	2399076	2019-05-05 01:38:00	2	1	1	0	0	0	0	4.04	DS	3.18	6	0	2019-05-05 01:38:11
9190	2672021	2019-05-05 01:35:37	2	1	1	0	0	0	0	4.08	DS	3.16	6	0	2019-05-05 01:35:48
9189	2399076	2019-05-05 01:31:38	2	1	1	0	0	0	0	4.04	DS	3.17	7	0	2019-05-05 01:31:49
9188	2672021	2019-05-05 01:29:39	2	1	1	0	0	0	0	4.07	DS	3.19	6	0	2019-05-05 01:29:50
9187	2399076	2019-05-05 01:25:14	2	1	1	0	0	0	0	4.05	DS	3.23	6	0	2019-05-05 01:25:25
9186	2672021	2019-05-05 01:23:41	2	1	1	0	0	0	0	4.08	DS	3.35	6	1	2019-05-05 01:23:51
9185	2399076	2019-05-05 01:18:50	2	1	1	0	0	0	0	4.04	DS	3.34	6	0	2019-05-05 01:19:01
9184	2672021	2019-05-05 01:17:43	2	1	1	0	0	0	0	4.06	DS	0.63	6	1	2019-05-05 01:17:54
9183	2399076	2019-05-05 01:12:29	2	1	1	0	0	0	0	4.06	DS	0.13	6	0	2019-05-05 01:12:40
9182	2399076	2019-05-05 01:09:57	2	1	1	0	0	0	0	4.05	DS	0.40	6	0	2019-05-05 01:10:08
9181	2672021	2019-05-05 01:05:38	2	1	1	0	0	0	0	4.04	DS	0.42	6	0	2019-05-05 01:05:49
9180	2672021	2019-05-05 01:04:31	2	1	1	0	0	0	0	4.08	DS	0.14	6	0	2019-05-05 01:04:42
9179	2399076	2019-05-05 01:03:02	2	1	1	0	0	0	0	4.05	DS	0.29	6	0	2019-05-05 01:03:13
9178	2672021	2019-05-05 00:58:40	2	1	1	0	0	0	0	4.06	DS	0.75	6	0	2019-05-05 00:58:51
9177	2399076	2019-05-05 00:51:03	2	1	1	0	0	0	0	4.05	DS	0.32	6	0	2019-05-05 00:51:14
9176	2672021	2019-05-05 00:28:42	2	1	1	0	0	0	0	4.06	DS	0.31	6	0	2019-05-05 00:28:53
9175	2399076	2019-05-05 00:24:41	2	1	1	0	0	0	0	4.04	DS	0.38	6	0	2019-05-05 00:24:52
9174	2672021	2019-05-05 00:23:44	2	1	1	0	0	0	0	4.04	DS	0.38	6	0	2019-05-05 00:23:55

Database: To store all the incoming sensor data from DAD

Variables (*Please select up to 5)

× Live - PPFD

PPFD – Photosynthetic Photon Flux Density $\mu\text{Mol}/\text{m}^2/\text{s}$.



Location/Device (*Please select up to 3)

× TGS TRIAL

× TGS TRIAL

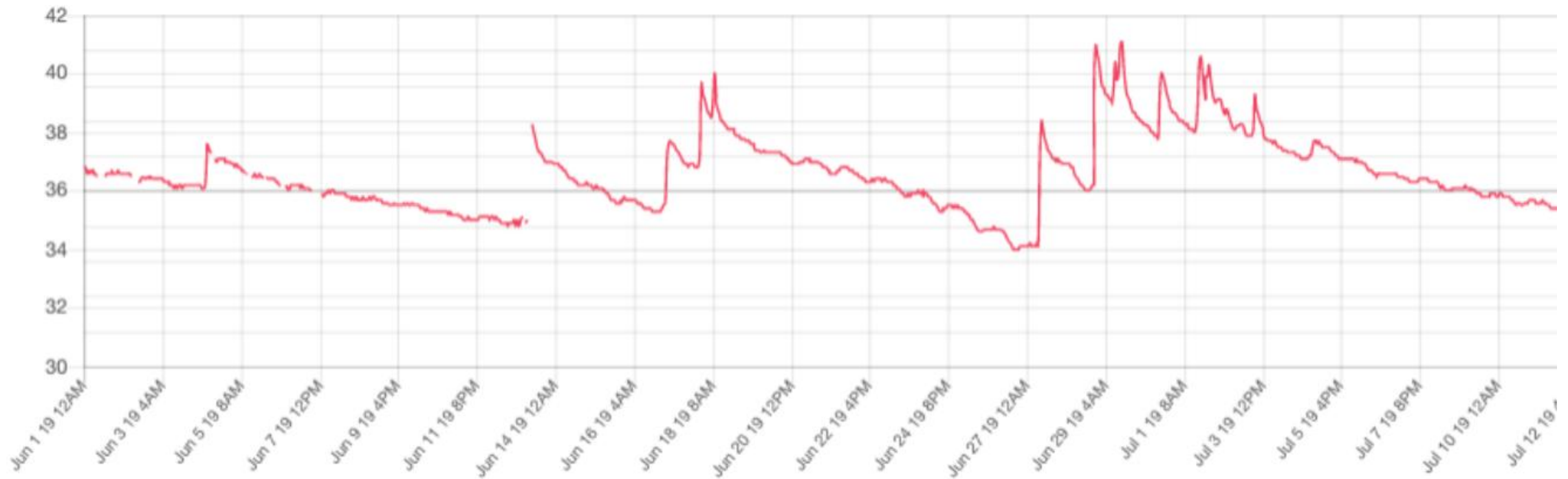
× TGS TRIAL

Analytics

Variables (*Please select up to 5)

× Live - VWC 1

Volume metric Water Content



Location/Device (*Please select up to 3)

× 021 - Blk 1-G - Dnd#76 (lip)

Execute

Analytics

Variables (*Please select up to 5)

✖ Live - VWC1 ✖ Live - VWC2

Volume metric Water Content



Location/Device (*Please select up to 3)

✖ 496 - Blk 3-F1 - Dnd#80

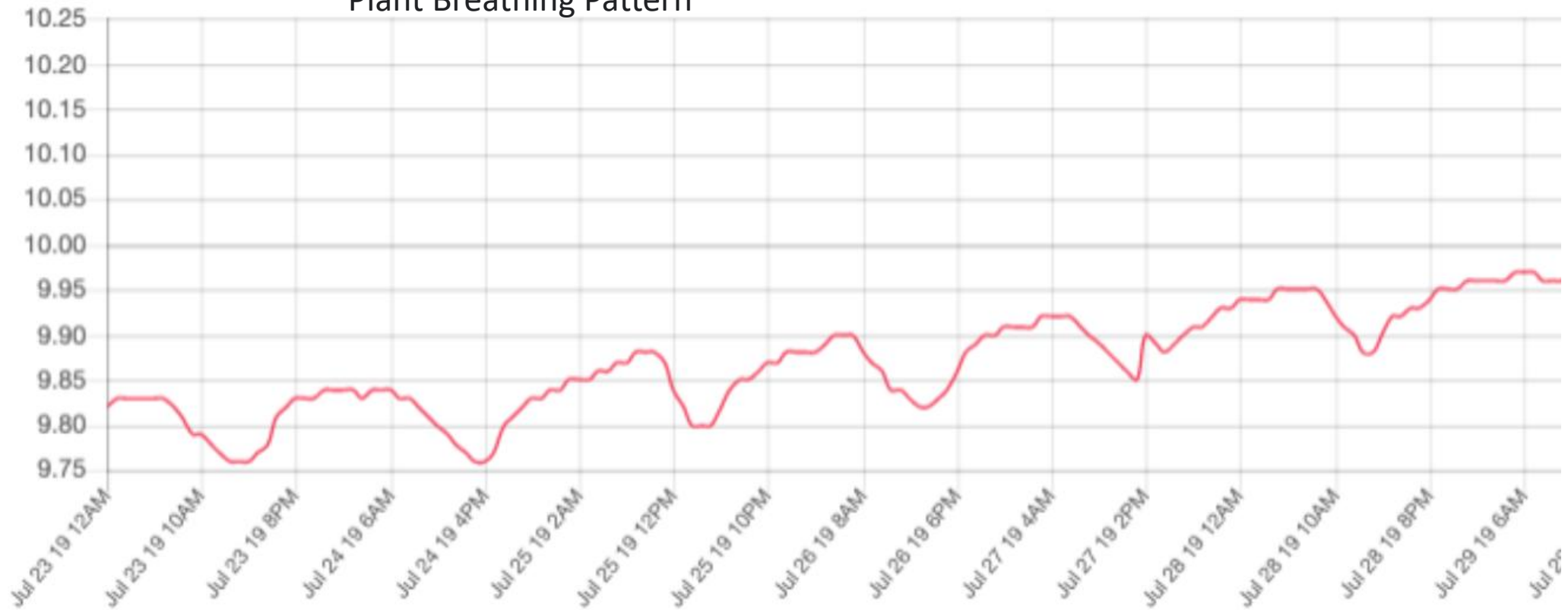
Execute

Clear

Variables (*Please select up to 5)

× Live - Stem Diameter #1

Plant Breathing Pattern



Location/Device (*Please select up to 3)

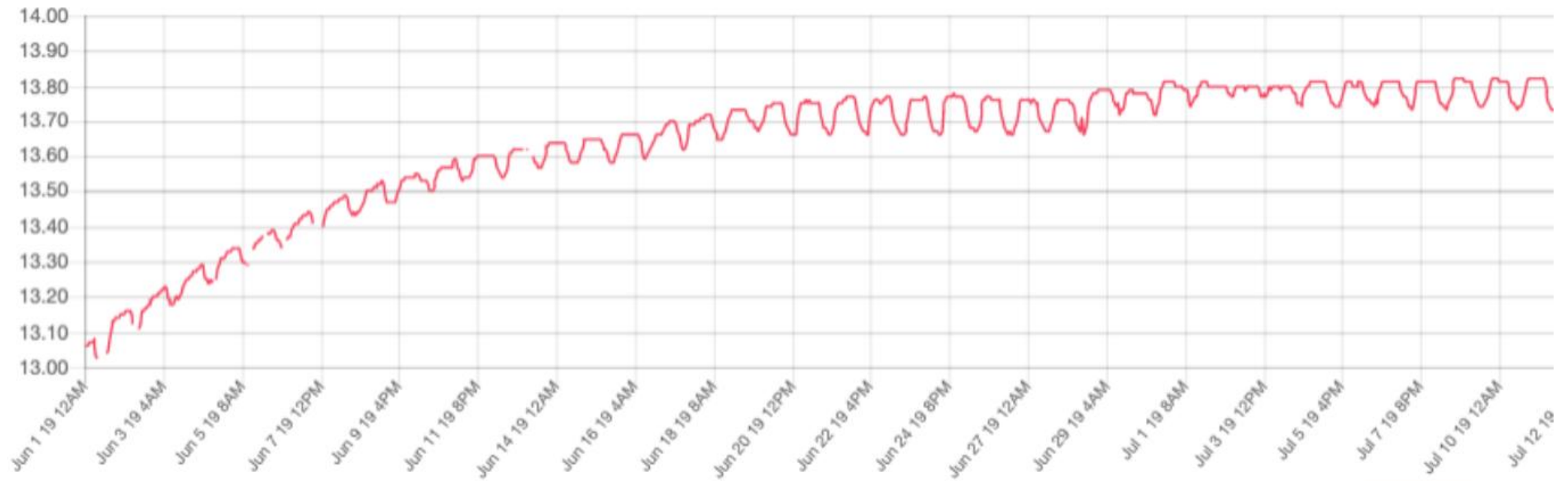
× 225 - Blk 8-A -Dnd#84 (lipless - with clip)

Plant Breathing Pattern

Analytics

Variables (*Please select up to 5)

× Live - Stem Diameter #1



Location/Device (*Please select up to 3)

× 021 - Blk 1-G - Dnd#76 (lip)

Execute

Daily Plant Breathing Pattern

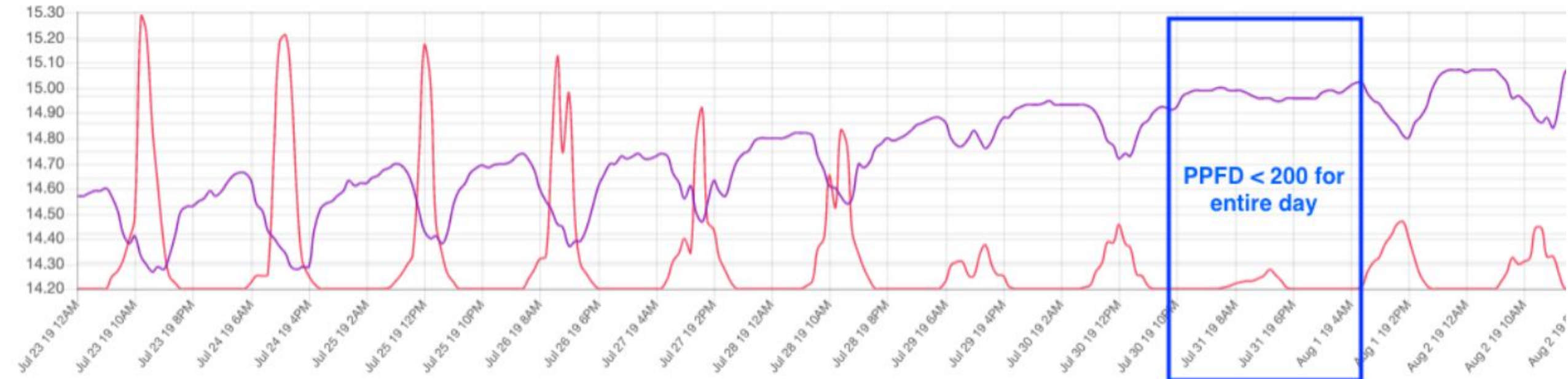
Analytics

Variables (*Please select up to 5)

Date Range

2019-07-23 - 2019

☒ Live - PPFD
 ☒ Live - Stem Diameter #1



Location/Device (*Please select up to 3)

☒ 492 - Blk 8-H - Dnd#83 (lipless - with clip)

Execute

Clear

Daily Plant Breathing Pattern

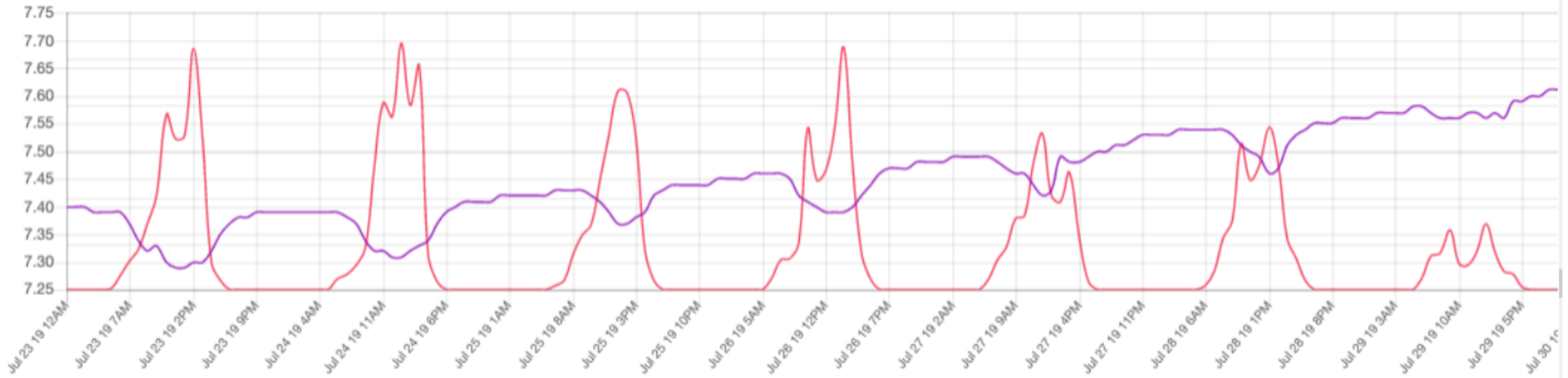
Analytics

Variables (*Please select up to 5)

☒ Live - PPFD
 ☒ Live - Stem Diameter #1

Date Range

2019-07-23 - 2019

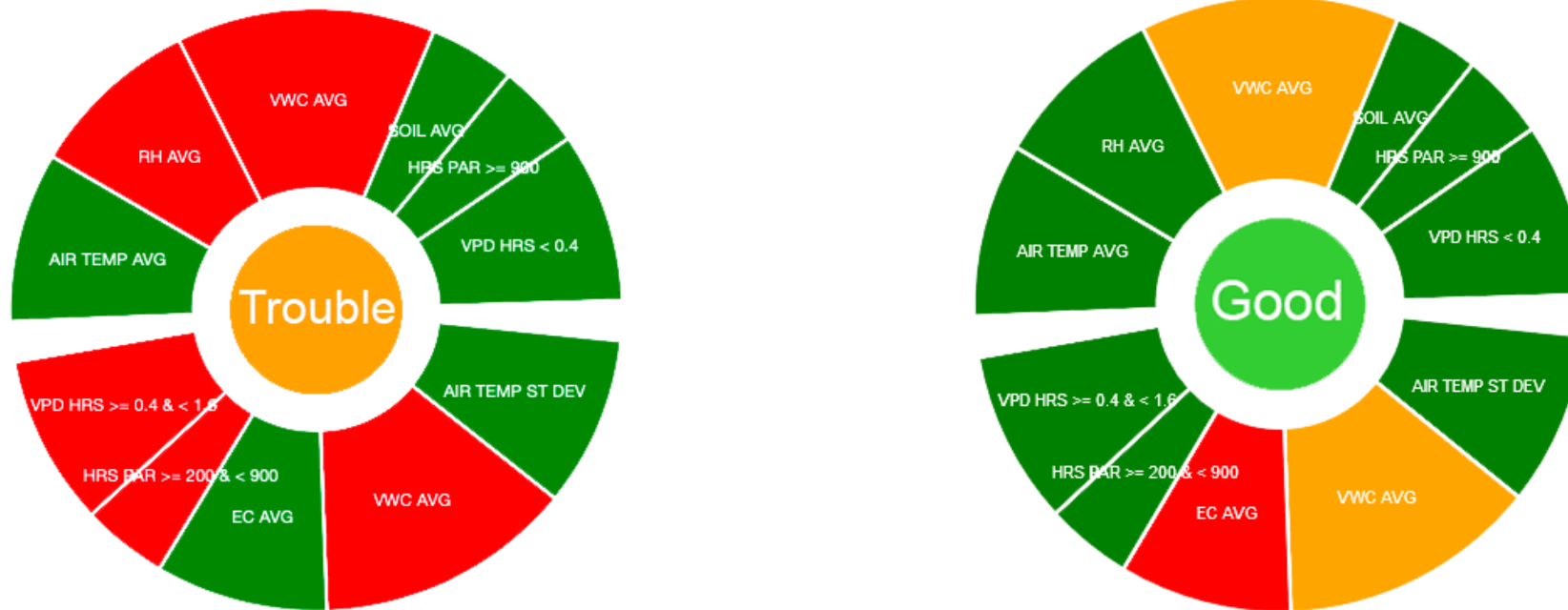


Stem breathing pattern

- Stem shrinks and grows based on the sun light effect(Photo syntheses)



Prediction – Plant current health condition based on various sensed Parameters



Smart Agri platform analyzes data and presents the gathered information in an insightful visual of crop health. This user-defined interface allows growers to decide which variables they would like to see. The Eye monitors environmental actors that are affecting crop health, various sectors of land, and crop's surroundings in order to ensure they are growing in optimal conditions.

Displays specific crop-health information that is derived from light, climate and soil data.

Gives growers a greater understanding of environmental aspects that need to be altered or kept the same in order to ensure an optimized yield.

Smart Irrigation – iStarter Product

Temperature Monitor Device -TRemo



Tracker T-Remo

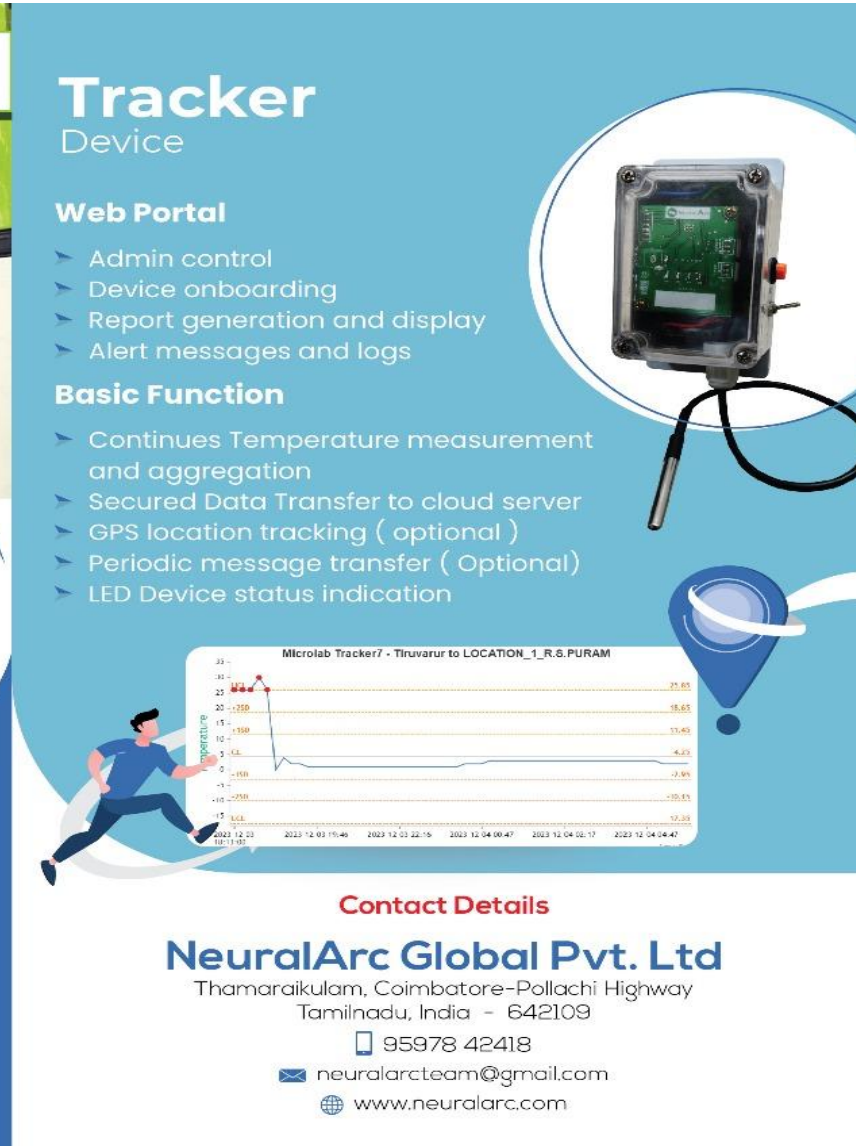
Temperature Monitoring Device

Web Portal

- Admin control
- Device onboarding
- Report generation and display
- Alert messages and logs

Basic Function

- Continues Temperature measurement
- Secured Data Transfer to cloud server
- LED Device status indication



Tracker Device

Web Portal

- Admin control
- Device onboarding
- Report generation and display
- Alert messages and logs

Basic Function

- Continues Temperature measurement and aggregation
- Secured Data Transfer to cloud server
- GPS location tracking (optional)
- Periodic message transfer (Optional)
- LED Device status indication

Contact Details

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 🌐 www.neuralarc.com

Clients

80 +

Blood Bank Software



Blood Bank Software

- Purchase bill
- Empty Bag stock
- Donor Information
- Screening Report
- Blood Bag Stock
- Blood Request Information
- Branch Based Empty & Blood Bag Stock

Contact Us:

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 neuralarcteam@gmail.com
 www.neuralarc.com



Clients

3 +

Inventory Management Software

Total Assets

16,689

[View Details](#)

Total Assets Value

148

[View Details](#)

Assigned Assets

\$156K

[View Details](#)

Unassigned Assets

16,689

[View Details](#)

Monthly Purchase Cost

148

[View Details](#)

Monthly Assets Used

\$156K

[View Details](#)

Cost of Assets Per Customer

[View Details](#)

Assets Depreciation Cost

[View Details](#)

Assets Return

[View Details](#)

Clients

2 +

Course Management Software-TRemo

[Competition](#) [Contact Us](#) [Gallery](#)

IGNITING YOUNG MINDS
Free Coaching



Exams only for Govt School students

Course :NMMS
Exam Month : Oct/Nov
Standard : 8th std

Courses :TRUST
Exam Month : Oct/Nov
Standard : 9th std

Clients

1 +

Digital Starter

Digital Starter for 3 Phase Motor

Motor Protection

- High Voltage Protection
- Low voltage Protection
- Over Load Protection
- Dry Run Protection
- Single Phase Failure Preventor

Display

- Voltage / Current value indication
- Cumulative current utilization
- R/Y/B phase status indication
- Power status indication
- Motor status indication
- Operational mode (Manual / Auto) indication



Input Supply 415 volt AC
Fully Load Current 10A

HP 0.5 to 5

Mobile App for Remote Operation



- Motor Start / Stop
- Cyclic Timer
(ON the Motor for some duration and restart the motor after some duration automatically)
- Timer
(ON the motor from now and run it for a set duration and then stop automatically)
- Scheduled Timer
(On daily or weekly basis, start and stop at a particular time automatically)
- Auto Start
(Start whenever power available and stop on Dry run)
- Solenoid Valve based water feeder control
- Water tank Level Automatic control



Internship Course



NeuralArc

Thamaraikulam,
Coimbatore-Pollachi Highway,
Tamilnadu, India - 642109.



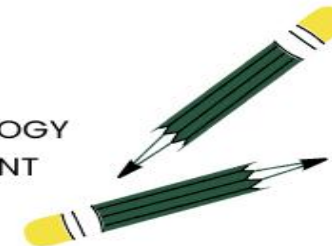
Colleges

4 +

NEURALARC GLOBAL PRIVATE LIMITED INTERNSHIP COURSES

- ✓ PYTHON
- ✓ INTERNET OF THINGS
- ✓ FULLSTACK WEB DEVELOPMENT
- ✓ WORDPRESS
- ✓ POWERBI
- ✓ C LANGUAGE

- ✓ INFORMATION TECHNOLOGY
- ✓ MOBILE APP DEVELOPMENT
- ✓ AI / ML
- ✓ DATA ANALYTICS
- ✓ 4G & 5G NETWORK ARCHITECTURE



How to Help him ?



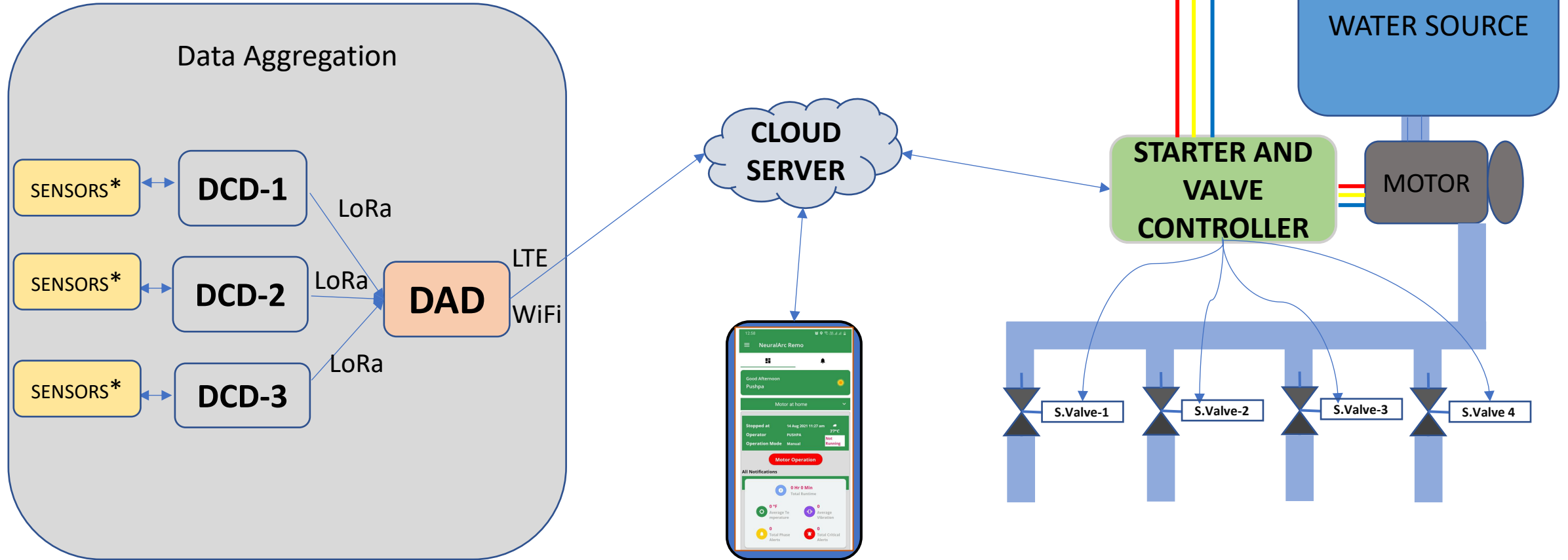
"How can I ensure growth
when everything is
constantly changing?"

**OPTIMIZE YOUR GROW
OPERATIONS IN A SINGLE
PLATFORM USING:**

1. Data Collection
2. Data Processing
3. Health vs. Hazard Insights



End To End Smart Irrigation



Sensors* - NPK sensor, Soil temperature & Moisture, Ambient temperature & humidity and Light sensor

Smart Irrigation - iStarter Features

- **Mobile App for Remote Operation**

- ON / OFF
- Cyclic Timer (ON the Motor for some duration and restart the motor after some duration automatically)
- Timer (ON the motor from now and run it for a set duration and then stop automatically)
- Scheduled Timer (On daily or weekly basis, start and stop at a particular time automatically)
- Auto Start (Start whenever power available and stop on Dry run)
- Solenoid Valve based water feeder control
- Water tank Level Automatic control

- **Motor Protection**

- High Voltage Protection
- Low voltage protection
- Over load protection
- Dry run protection
- Single Phase failure preventor

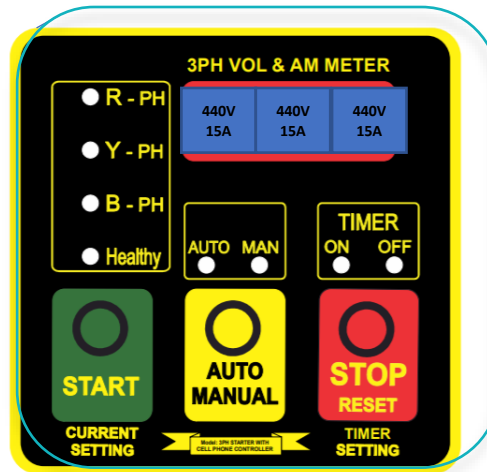
- **Display**

- Voltage / Current value indication
- Cumulative current utilization
- R/Y/B phase status indication
- Power status Indication
- Motor status indication
- Operational mode(Manual / Auto) indication

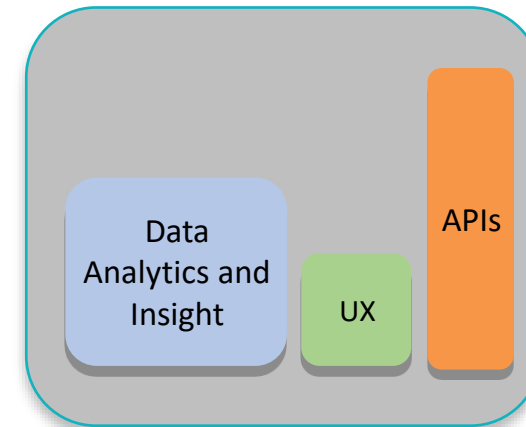
- **Sensor based Automation**

- Soil Moisture based automatic irrigation

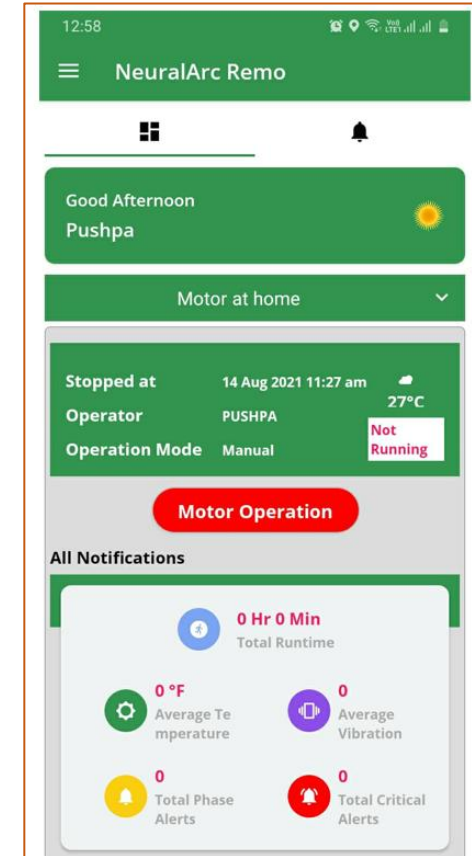
Physical iStarter

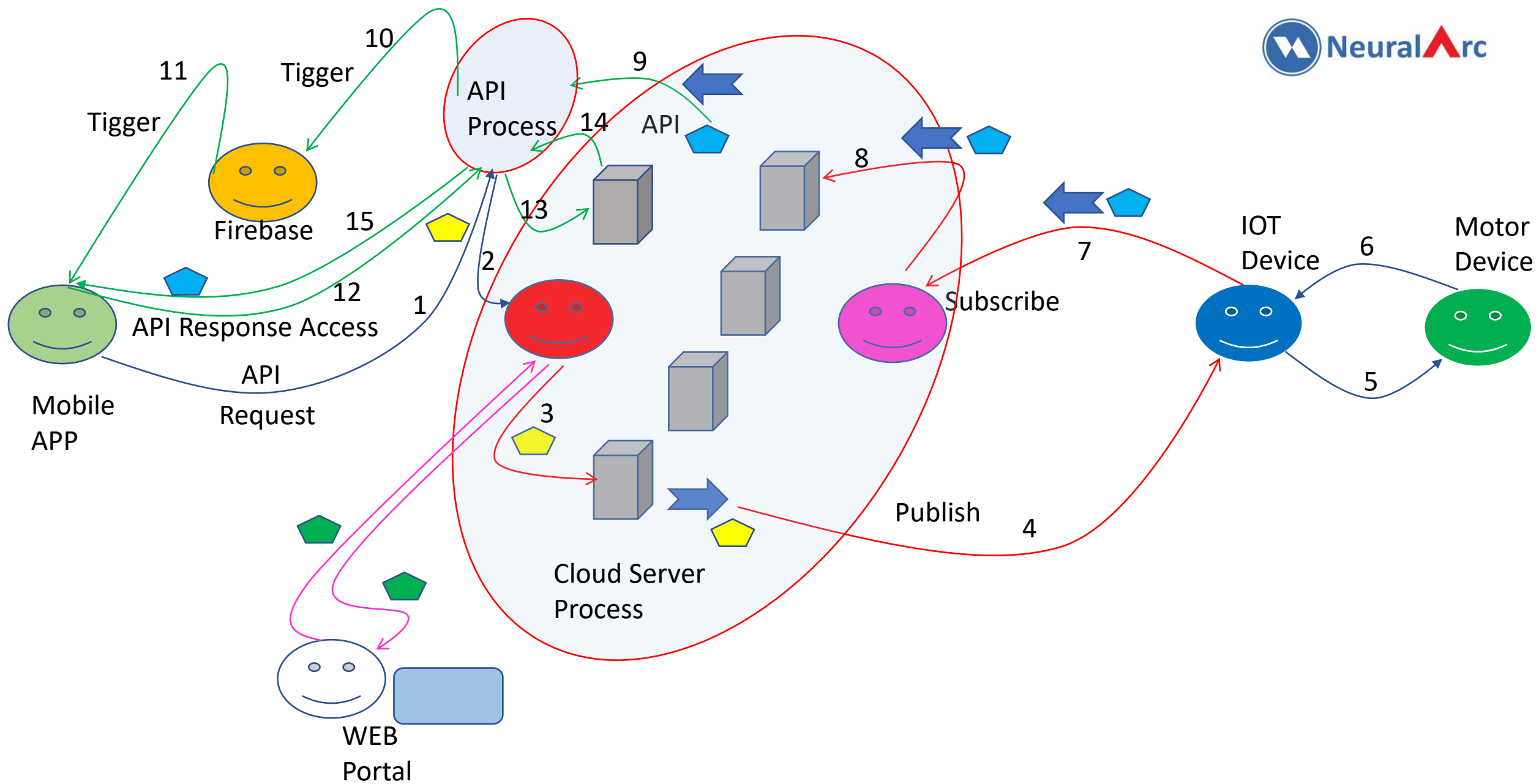


Cloud Server

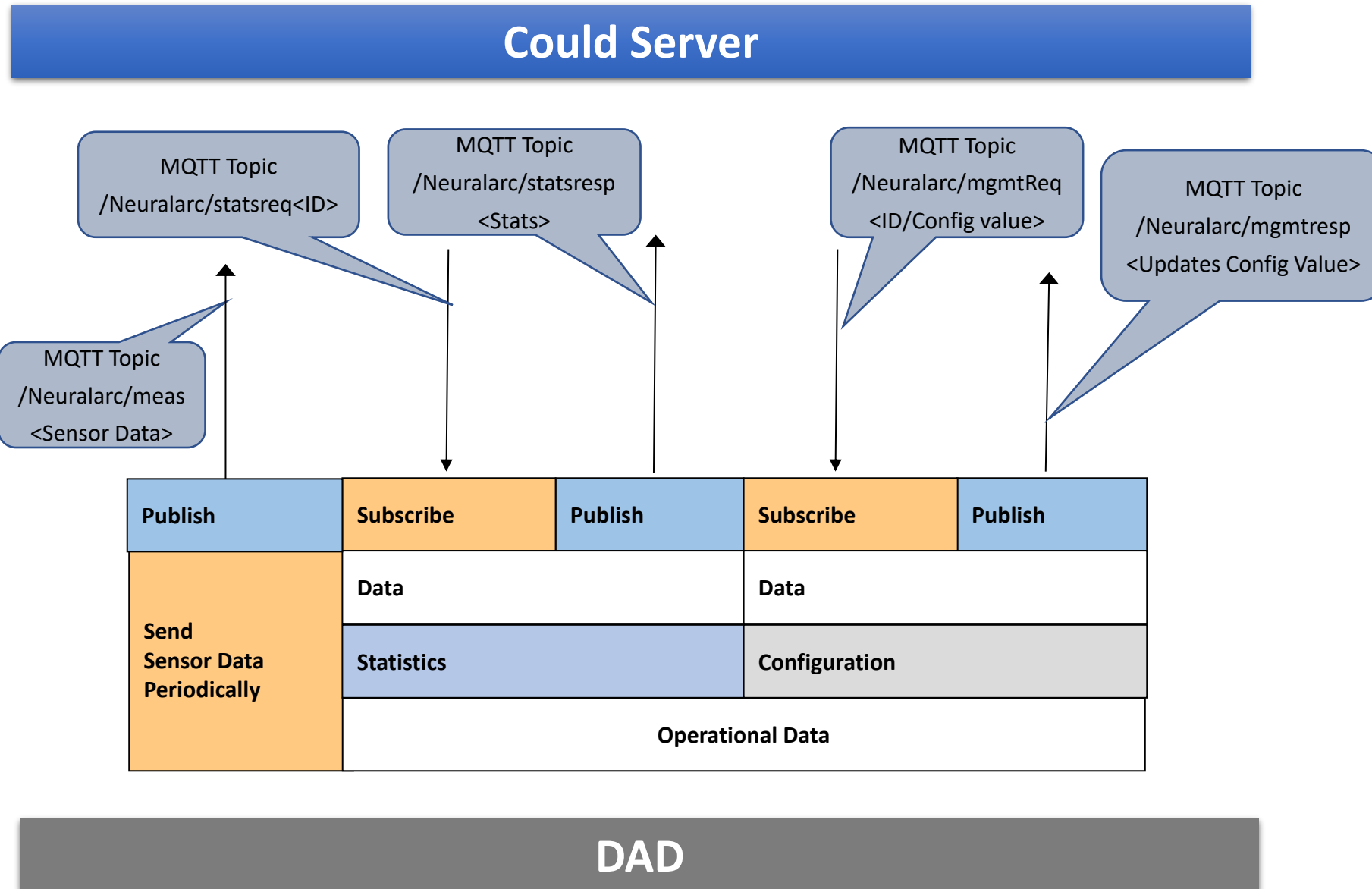


Mobile Application

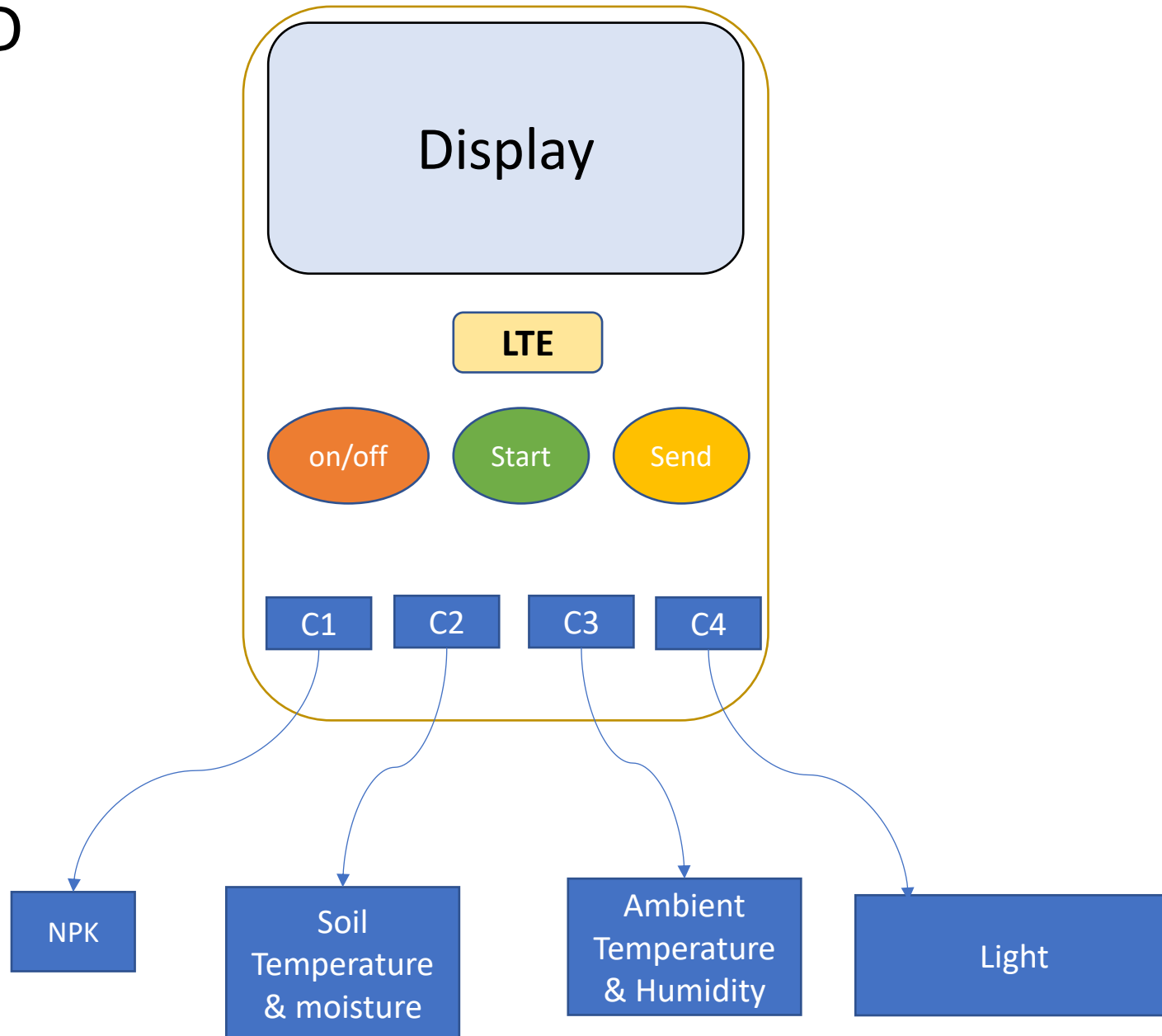




DAD <-> Cloud Server MQTT Protocol



Handheld DCD



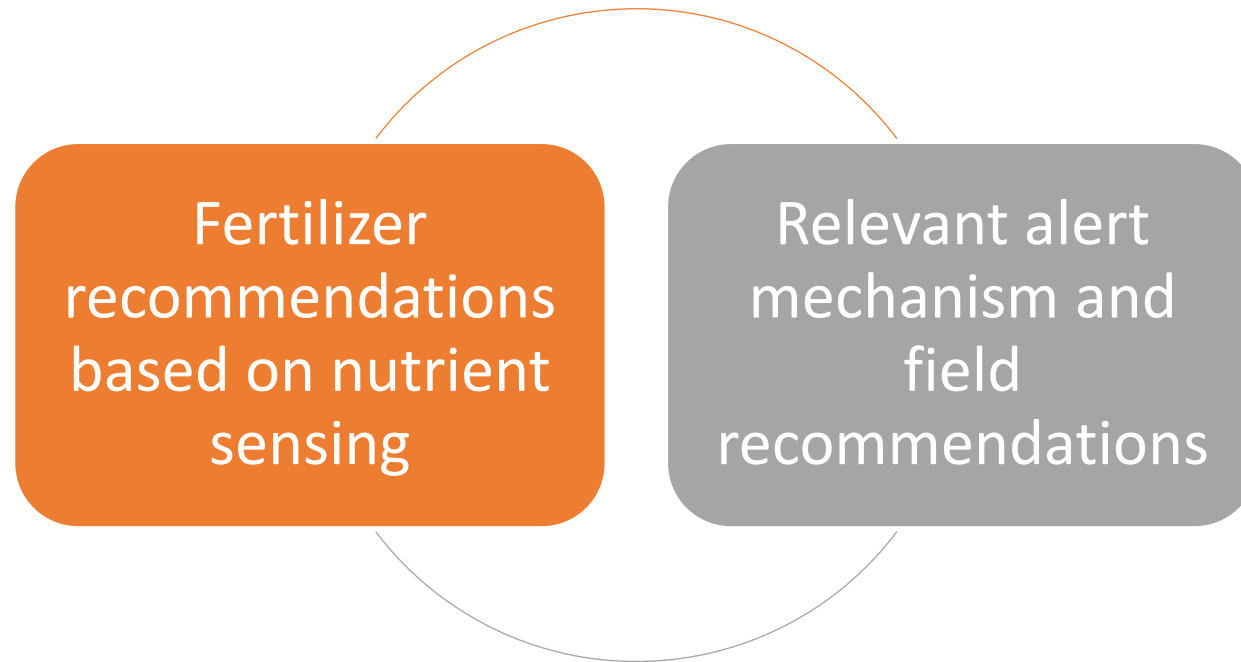
Support system

NeuralArc Commitments

- On-site support/training during setup
- Make timely adjustments/fixes
- Available to answer questions when off-site/during regular Project reviews
- Involve Customer in hardware/software roadmap planning
- Trouble-shoot hardware and software issues
- Document user input/complaints/suggestions

Sustainable Soil Management - Nutrisoil

Fertilizer Recommendation



Data Analytics and ML/AI Tools

Delphine Data Analytics Tool

Delphine Diagnostic Inc (USA) AI Tool

- Pathogen Predictions – Faster In-sight delivery to Patients
- ML/AL based medical advise Tool to General Doctors for effective treatment and cost control

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Activities GVim Mar 22 13:06

gmm-handler.c (~/Ranjana/openSgs/src/amf) - GVIM

File Edit Tools Syntax Buffers Window Help

s/u/n/m/register.cpp a/t/SL_tx.cc a/k/SL_kb.cc s/ue.cpp ../o/s/a/gmm-handler.c

```
888     int portno =      80;
889     //if 0
890     char *host =      "agri.neuralarc.com";
891     //char *host =      "agri.neuralarc.com/mobile_app/php_test/";
892     //char *host =      "https://agri.neuralarc.com/mobile_app/php_test/attach.php";
893     char *message_fmt = "POST / table=23&rid=23&cid=43&mdate=12-02-2021&t1=0.2&rhl=22&lux=12&wtlux=21&st1=12&p1=21&ec1=11&d1=23&BV=33&BC=22&SV=11&RS=11&st2=21&p2=12&ec2=12 HTTP/1.0\r\n\r\n";
894     //char *message_fmt = "POST /q=%s&oq=%s&aqs=%s&client=%s&sourceid=%s&ie=%sHTTP/1.0\r\n\r\n"; //for google/search
895
896     struct hostent *server;
897     struct sockaddr_in serv_addr;
898     int sockfd, bytes, sent, received, total;
899     char message[1024], response[4096];
900
901     /* fill in the parameters */
902     //sprintf(message,message_fmt,argv[1],argv[2],argv[3],argv[4],argv[5],argv[6],argv[7],argv[8],argv[9]);
903     printf("Request:\n%s\n",message_fmt);
904
905     /* create the socket */
906     sockfd = socket(AF_INET, SOCK_STREAM, 0);
907     if (sockfd < 0) error("ERROR opening socket");
908
909     /* lookup the ip address */
910     server = gethostbyname(host);
911     if (server == NULL) error("ERROR, no such host");
912
913     /* fill in the structure */
914     memset(&serv_addr,0,sizeof(serv_addr));
915     serv_addr.sin_family = AF_INET;
916     serv
917     memcp
918
919     /* co
920     if (c
921
922     error("ERROR connecting");
```

08:22 [Microphone Icon] [Screen Share Icon] [More Icon] [Hand Icon] [Chat Icon] [Participants Icon] [End Call Icon]

893,33-40 68%

Ranjana

MA

VS

SK

PK

R

Vaibhav Shivaji Sable

Subha Krishna

Pooja Koti

Dinesh Kumar Srinivasa M...

Ranjana



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Activities XTerm Mar 22 13:11

ranjana@ranjana-ThinkPad-E470: ~/Ranjana/UERANSIM/app/build

ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T...

```
***** Welcome to Cienra IoT App*****  
*Operations.....  
* Send SMS  
* Send file  
* Broadcast Message  
* sh <any-linux-command> ==> any shell command  
***** Call us at 9886212362 for any clarification *****  
  
:iAPP>>Server IP Address: 127.0.0.1  
  
:iAPP>>loginrequest  
  
:iAPP>> 
```

log_app

kb->app: handle_kb_msg=401, ue_id=0
app: handle_kb_msg
13:09:50.277794:App->tx:N_IOT_LOGIN_REQ:401
AMF:app: handle_rx_msg=402, ue_id=901700000001001
APP:Received +ve N_IOT_LOGIN_RESP

AMF:app: handle_rx_msg=404, ue_id=0
APP:Received N_IOT_REGISTER_ACCEPT

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R



1:11 PM
3/22/2022



Activities Terminal Mar 22 13:11

ranjana@ranjana-ThinkPad-E470: ~/Ranjana/UEANSIM/cmake-build-debug

ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T...

UE Process Instance created

My IP Address: 0.0.0.0
(defaultServer:Local)-Net present,If Remote:1 <192.168.11.74>, else local:0:

Server IP Address: 127.0.0.1[2022-03-22 13:10:46.225] [nas] [info] UE switches to state [MM-DEREGISTERED/PLMN-SEARCH]
[2022-03-22 13:10:46.284] [rrc] [debug] New signal detected for cell[1], total [1] cells in coverage
[2022-03-22 13:10:51.730] [nas] [info] UE switches to state [MM-DEREGISTERED/NO-CELL-AVAILABLE]
[2022-03-22 13:10:52.270] [nas] [info] Selected plmn[901/70]
[2022-03-22 13:10:52.270] [rrc] [info] Selected cell plmn[901/70] tac[1] category[SUITABLE]
[2022-03-22 13:10:52.271] [nas] [info] UE switches to state [MM-DEREGISTERED/PS]
[2022-03-22 13:10:52.286] [nas] [info] UE switches to state [MM-DEREGISTERED/NORMAL-SERVICE]
[2022-03-22 13:10:52.286] [nas] [debug] Initial registration required due to [MM-DEREG-NORMAL-SERVICE]
[2022-03-22 13:10:52.288] [nas] [debug] UAC access attempt is allowed for identity[0], category[MO_sig]
[2022-03-22 13:10:52.288] [nas] [debug] Sending Initial Registration
[2022-03-22 13:10:52.296] [nas] [info] UE switches to state [MM-REGISTER-INITIATED]
[2022-03-22 13:10:55.195] [nas] [debug] Authentication Request received
[2022-03-22 13:10:57.195] [nas] [debug] Security Mode Command received
[2022-03-22 13:10:57.195] [nas] [debug] Selected integrity[2] ciphering[0]
[2022-03-22 13:10:58.196] [nas] [debug] Registration accept received
[2022-03-22 13:10:58.199] [nas] [info] UE switches to state [MM-REGISTERED/NORMAL-SERVICE]
[2022-03-22 13:10:58.199] [nas] [info] GUTI recieved in registration accept
[2022-03-22 13:10:58.199] [nas] [info] PLMN MCC = 901, MNC = 70
[2022-03-22 13:10:58.199] [nas] [info] AMFRegion/Set/Pointer: 2/1/0
[2022-03-22 13:10:58.199] [nas] [info] TMSI = DA005875
[2022-03-22 13:10:58.201] [nas] [debug] N_GNB_EMM_CONTEXT_EXIST:
[2022-03-22 13:10:58.201] [nas] [debug] CM State changed to ---> CM_CONNECTED
[2022-03-22 13:10:58.201] [nas] [debug] Sending Registration Complete
[2022-03-22 13:10:58.201] [nas] [info] Initial Registration is successful
[2022-03-22 13:10:58.202] [nas] [debug] Sending PDU Session Establishment Request
[2022-03-22 13:10:58.202] [nas] [debug] UAC access attempt is allowed for identity[0], category[MO_sig]
[2022-03-22 13:10:59.201] [nas] [debug] Configuration Update Command received
[2022-03-22 13:11:02.202] [nas] [debug] PDU Session Establishment Accept received
[2022-03-22 13:11:02.202] [nas] [info] PDU Session establishment is successful PSI[1]
[2022-03-22 13:11:02.202] [app] [error] TUN interface could not be setup. Permission denied. Please run the UE with 'sudo'

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Activities Terminal Mar 22 13:11

ranjana@ranjana-ThinkPad-E470: ~/Ranjana/openSgs

ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T... ranjana@ranjana-T...

```
03/22 13:10:56.494: [sbi] WARNING: Unknown tm_gmtoff[19800:19800] (./lib/sbi/conv.c:273)
03/22 13:10:57.529: [app] WARNING: Try to discover [UDM] (./lib/sbi/path.c:110)
03/22 13:10:57.579: [amf] INFO: [f0be5334-a9a4-41ec-bbf9-aff94b8fed1f] (NF-discover) NF registered (./src/amf/nnrf-handler.c:342)
03/22 13:10:57.579: [amf] INFO: [f0be5334-a9a4-41ec-bbf9-aff94b8fed1f] (NF-discover) NF Profile updated (./src/amf/nnrf-handler.c:402)
03/22 13:10:57.901: [sbi] WARNING: Unknown tm_gmtoff[19800:19800] (./lib/sbi/conv.c:273)
03/22 13:10:58.494: [gmm] INFO: [imsi-901700000001001] Registration complete (./src/amf/gmm-sm.c:1067)
03/22 13:10:58.494: [amf] INFO: [imsi-901700000001001] Configuration update command (./src/amf/nas-path.c:389)
03/22 13:10:58.494: [gmm] INFO: UTC [2022-03-22T07:40:58] Timezone[0]/DST[0] (./src/amf/gmm-build.c:505)
03/22 13:10:58.494: [gmm] INFO: LOCAL [2022-03-22T13:10:58] Timezone[19800]/DST[0] (./src/amf/gmm-build.c:510)
03/22 13:10:59.518: [nas] INFO: Decode UL_NAS_TRANSPORT payload container type 1
(./lib/nas/5gs/decoder.c:2163)
03/22 13:10:59.518: [gmm] INFO: ***** In 5GMM Msg Event Handling on AMF ***** (./src/amf/gmm-sm.c:123)
03/22 13:10:59.518: [gmm] INFO: *****UL NAS Transport***** (./src/amf/gmm-sm.c:332)
03/22 13:10:59.542: [amf] INFO: [Added] Number of AMF-Sessions is now 1 (./src/amf/context.c:2048)
03/22 13:10:59.542: [gmm] INFO: UE SUPI[imsi-901700000001001] DNN[internet] S_NSSAI[SST:1 SD:0xfffff] (./src/amf/gmm-handler.c:1154)
03/22 13:10:59.542: [sbi] WARNING: Unknown tm_gmtoff[19800:19800] (./lib/sbi/conv.c:273)
03/22 13:10:59.742: [smf] INFO: [Added] Number of SMF-UEs is now 1 (./src/smf/context.c:793)
03/22 13:10:59.831: [smf] INFO: [Added] Number of SMF-Sessions is now 1 (./src/smf/context.c:2610)
03/22 13:11:00.037: [app] WARNING: Try to discover [PCF] (./lib/sbi/path.c:110)
03/22 13:11:00.087: [smf] WARNING: [f0bed9c6-a9a4-41ec-a296-db810e347482] (NF-discover) NF has already been added (./src/smf/nnrf-handler.c:289)
03/22 13:11:00.162: [smf] INFO: [f0bed9c6-a9a4-41ec-a296-db810e347482] (NF-discover) NF Profile updated (./src/smf/nnrf-handler.c:330)
03/22 13:11:00.304: [app] WARNING: Try to discover [BSF] (./lib/sbi/path.c:110)
03/22 13:11:00.390: [pcf] WARNING: [f0be7e90-a9a4-41ec-9a20-7b38ff1f25c5] (NF-discover) NF has already been added (./src/pcf/nnrf-handler.c:291)
03/22 13:11:00.443: [pcf] INFO: [f0be7e90-a9a4-41ec-9a20-7b38ff1f25c5] (NF-discover) NF Profile updated (./src/pcf/nnrf-handler.c:346)
03/22 13:11:00.696: [smf] INFO: UE SUPI[imsi-901700000001001] DNN[internet] IPv4[10.45.0.4] IPv6[] (./src/smf/npcf-handler.c:492)
03/22 13:11:00.758: [upf] INFO: [Added] Number of UPF-Sessions is now 1 (./src/upf/context.c:160)
03/22 13:11:00.899: [upf] INFO: UE F-SEID[CP:0x3 UP:0x3] APN[internet] PDN-Type[1] IPv4[10.45.0.4] IPv6[] (./src/upf/context.c:361)
03/22 13:11:01.097: [app] WARNING: Try to discover [AMF] (./lib/sbi/path.c:110)
03/22 13:11:01.129: [smf] WARNING: [f0be95f6-a9a4-41ec-a49b-b5e6978d0cbb] (NF-discover) NF has already been added (./src/smf/nnrf-handler.c:289)
03/22 13:11:01.154: [smf] INFO: [f0be95f6-a9a4-41ec-a49b-b5e6978d0cbb] (NF-discover) NF Profile updated (./src/smf/nnrf-handler.c:330)
03/22 13:11:01.222: [amf] INFO: *****N1N2MessageTransfer*****
(./src/amf/namf-handler.c:62)
```

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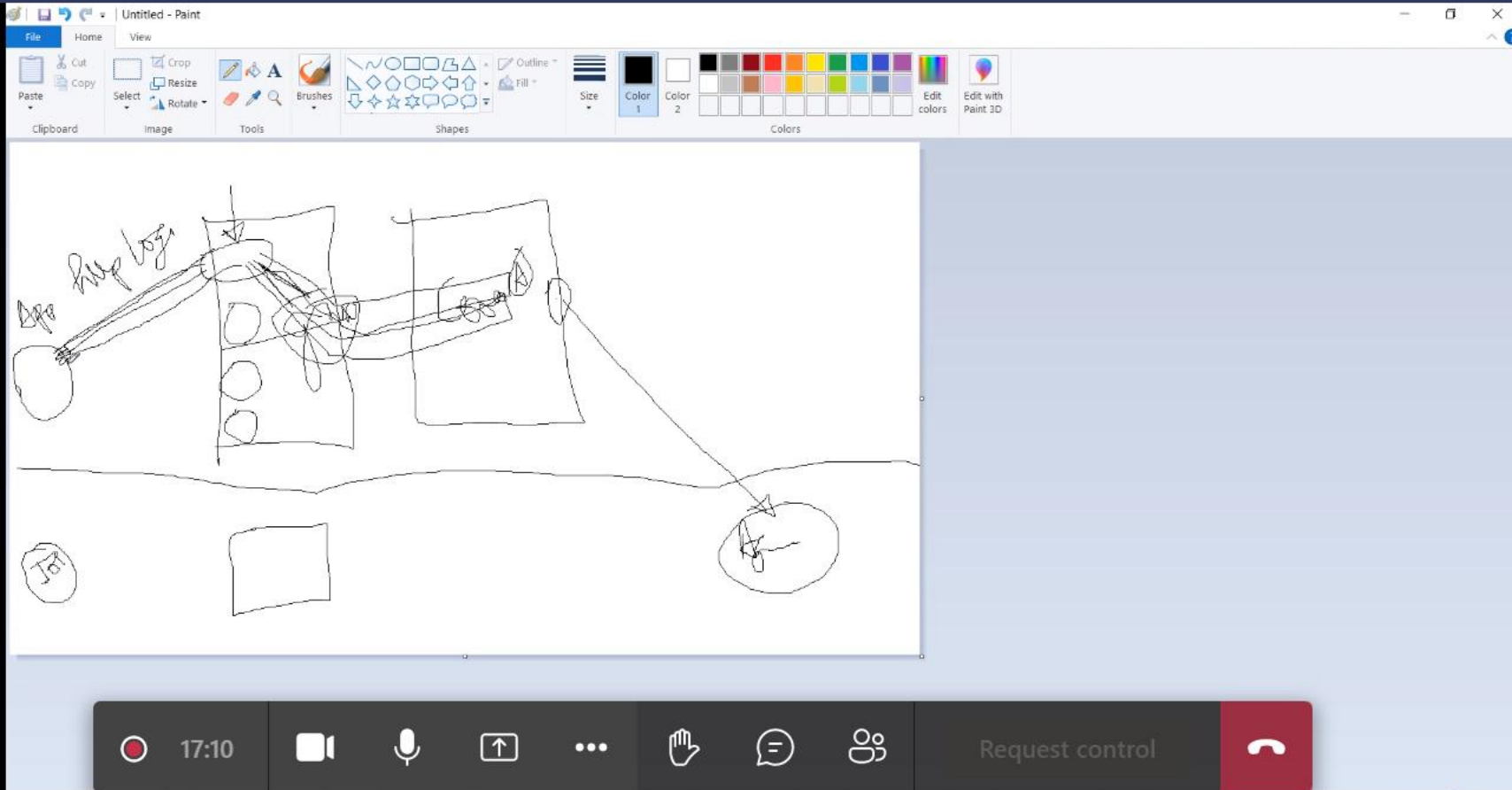


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